

Capstone Proposal: Retrieval, Ranking, and Trustworthy Offline Evaluation

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Track	AI Engineering
Github Repo	https://github.com/lephgrueber/ai-capstone-trustworthy-recsys
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Project Charter

Problem Statement

Recommendation systems have to solve two linked problems. First, users need relevant results quickly from catalogs that are too large to score exhaustively, so ranking has to happen in stages under a hard latency budget. Second, once a system is live, improving it becomes hard to measure. Logged interaction data is biased by the model that produced it: users only interacted with what they were shown, and top positions receive more attention regardless of relevance. Teams can therefore end up A/B testing many candidate models because they do not trust their offline estimates, limiting how quickly the recommendation experience can improve.

This project builds the two-stage system and then addresses the measurement problem directly: can we estimate a new ranker's performance from biased logs accurately enough to decide what's worth testing live? The system will be implemented in one concrete recommendation domain and exposed through a consumer-facing application, so the final product demonstrates both the user experience and the retrieval, ranking, and evaluation infrastructure behind it.

Target Users and Context

The primary user is an end user trying to discover relevant items from a large catalog. The user interacts with a recommendation application, establishes or expresses preferences, receives a ranked set of recommendations, and can refine those preferences when the initial results are not satisfactory.

The specific domain will be selected after the initial dataset audit. Candidate datasets include MovieLens and Amazon Reviews 2023, but the choice will be driven by whether the data contains sufficiently deep user histories, item metadata, timestamps, and labels to support retrieval, reranking, and meaningful evaluation.

Value Proposition

For the user, the system reduces the effort required to find relevant items in a large catalog by combining long-term behavioral signals with the preferences expressed during the current interaction. Success means that useful recommendations appear quickly and that changing or refining a preference produces predictably different results.

The project also demonstrates how candidate ranking policies can be evaluated offline before being considered for online experimentation.

End Deliverable

The final deliverable will be an end-to-end consumer-facing recommendation system built around the two-stage retrieval and ranking architecture. The system will retrieve a candidate set using the two-tower ANN pipeline, rerank those candidates with LambdaRank, and present the resulting recommendations through a user-facing interface.

The project will also include the offline evaluation harness for comparing direct method, IPS, SNIPS, and doubly robust estimates against observed policy performance, along with the retrieval, ranking, latency, and estimator-reliability results.

The user-facing interface will support natural-language requests and follow-up refinement through an LLM-based orchestration layer that queries the recommendation system rather than generating recommendations directly. MCP-based tool access to the recommendation/search backend is a stretch implementation if time permits.

The recommendation backend will contain three primary stages:

1. **Retrieval.** Two-tower model with separate user and item encoders mapping to a shared embedding space, scored by dot product. Item embeddings are precomputed into an ANN index (FAISS/HNSW) so retrieval is a vector lookup rather than a model forward pass. Trained with sampled softmax over in-batch negatives, with the logQ correction for popularity-driven sampling bias, then mixed negatives adding hard examples mined from the index itself.
2. **Reranking.** The two towers can't learn user×item feature crosses by construction, which is the price of precomputation. The reranker recovers them: a LambdaRank model (LightGBM) over the top ~500 candidates, with cross features, position, and content signals.
3. **Offline evaluation.** Four estimators of policy value from logged data: direct method, IPS, self-normalized IPS, and doubly robust. Position bias and propensity correction will be handled using the information available in the selected evaluation dataset. Each estimator's accuracy will be checked against ground-truth policy value from randomized data where available, so the deliverable is not just “here are some estimates” but “here is which estimator you can trust, at what sample size, under what overlap conditions.”

Success Metrics

Offline Metrics

- **Retrieval:** Recall@k and a recall/latency Pareto curve across ANN configurations. Target: approximately Recall@100 $\geq$ 0.80 with p99 ANN retrieval latency below 50 ms, subject to adjustment after dataset benchmarking.
- **Ranking:** NDCG@10 and MRR against BM25-style and popularity baselines, on a temporal split rather than a random split.
- **OPE:** Error against observed policy value, estimator variance, and confidence-interval coverage, reported as a function of log size and policy overlap where supported by the data.
- **End-to-end performance:** Measure recommendation-serving latency from request to ranked results.

Product and user-experience goals

- The system should support the complete recommendation workflow through a user-facing interface rather than requiring notebook interaction.
- A predefined set of user scenarios will test whether the returned recommendations reflect the user's stated preferences and constraints.
- If conversational refinement is implemented, follow-up requests should update the recommendations consistently with the user's revised intent.

Feasibility, Constraints, and Ethics and Safety Plan

Tech Stack

Modeling

- Two-tower retrieval model, framework to be finalized
- LightGBM LambdaRank for reranking
- FAISS/HNSW for approximate nearest-neighbor search
- Implementation of direct method, IPS, SNIPS, and doubly robust evaluation

Backend / Recommendation System

- Precomputed item embeddings and served ANN index
- Reproducible training pipeline
- Offline evaluation harness

User Interface

- User-facing recommendation interface
- If conversational interaction is implemented, an LLM will interpret natural-language preferences and follow-up requests
- Optional Agentic Layer

- MCP/tool-based orchestration (possible stretch)

Development

- GitHub
- Basic CI smoke test

Data and Licensing Constraints

Dataset selection is the first implementation priority. Based on professor's feedback MovieLens and Amazon Reviews 2023 are the leading candidates for the consumer-facing recommendation system. The selected dataset must contain sufficiently rich user histories, item metadata, timestamps, and labels to support retrieval, reranking, meaningful baselines, and evaluation.

Open Bandit Dataset remains the leading dataset for the OPE component because it contains real logged bandit feedback with known propensities. If the selected recommendation dataset does not support the OPE experiments directly, the retrieval/ranking and OPE components will use separate datasets, as anticipated in the original proposal.

Before implementation, the team will document the licensing and usage constraints for the selected datasets and any external metadata or API sources used by the application.

Compute Budget and Tooling

The project is designed to use modest academic compute rather than large-scale model training. The primary compute requirements will come from training the two-tower retrieval model, building and benchmarking the ANN index, training the LambdaRank reranker, and running the OPE experiments.

The specific compute environment and budget will be finalized after the dataset is selected and its scale is known.

Risks and Mitigations

1. Dataset suitability

Real bandit datasets may have feature spaces that are too limited to support the retrieval and ranking system, while richer recommendation datasets may not contain the logged propensities required for OPE.

- Mitigation: Select the recommendation dataset based on the requirements of the consumer-facing system and use a separate bandit dataset for OPE if necessary. If

separate datasets are used, that boundary will be stated explicitly rather than treated as if both experiments came from the same system.

2. IPS variance

IPS variance may be large enough that some estimators are uninformative.

- Mitigation: Report estimator variance, confidence intervals, and performance as a function of log size and overlap. If an estimator becomes unreliable under poor overlap or limited data, that will be reported as a finding rather than treated as a project failure.

3. Scope

Retrieval, reranking, OPE, and the consumer-facing application create significant scope within the 14-week timeline.

- Mitigation: The OPE component and a working end-to-end recommendation flow are the priorities to protect. If the schedule slips, reranker sophistication and advanced features such as hard-negative mining, more sophisticated multi-turn interaction, or MCP-based orchestration will be reduced before the core evaluation work is cut.

Ethics and Safety Plan

Recommendation systems can reinforce popularity bias by repeatedly favoring already popular items. Where supported by the selected dataset, the team will examine the concentration of recommendations and whether the system disproportionately favors high-popularity items.

The project will use public datasets in accordance with their licensing and usage requirements. Any additional metadata sources or APIs used by the application will be reviewed for their applicable terms before integration.

If the LLM-based interaction layer is implemented, the model will be used to interpret user requests and interact with the recommendation system rather than serve as the source of recommendation candidates or item facts. Hallucinated or unsupported outputs will be included in failure testing.

Scope, Roles, and Timeline

In Scope

- Two-tower personalized retrieval
- ANN candidate retrieval
- LambdaRank reranking
- Recommendation baselines
- Consumer-facing recommendation interface
- Natural-language preference interpretation and multi-turn refinement through an LLM-based orchestration layer
- Reproducible training and evaluation pipeline
- DM, IPS, SNIPS, and DR evaluation
- Retrieval and ranking quality and latency measurement
- OPE reliability experiments
- End-to-end user-flow testing
- MCP-based tool access to the recommendation/search backend (stretch)

Out of Scope

- Production-scale deployment
- A statistically powered live A/B test
- A general-purpose recommendation platform spanning multiple domains
- Using an LLM as the underlying recommendation or ranking model

Timeline (14 weeks)

Weeks	Focus
1-2	Dataset/domain audit and selection; licensing review; temporal split; baseline implementation; initial UI mockup
3-5	Two-tower retrieval; sampled softmax; logQ correction
6-7	ANN index; latency benchmarking; hard-negative mining; begin user-interface prototype in parallel
8-9	LambdaRank reranker and cross features; integrate retrieval and reranking into an initial end-to-end recommendation flow
10-11	OPE harness implementing DM, IPS, SNIPS, and DR; integrate the consumer-facing interface and implement natural-language preference interpretation / follow-up refinement
12	OPE validation; sample-size and overlap experiments; end-to-end system

	evaluation
13	Failure analysis, ablations, final UI/system integration, packaging, and reproducibility testing; MCP-based tool access if time permits
14	Final documentation and technical report; finalize demo video; live showcase preparation and presentation

Roles and Responsibilities

Work Area	Responsibilities	Owner (s)
Data Retrieval	Dataset audit and selection, preprocessing, temporal split, retrieval data pipeline, initial two-tower setup	TBD
Ranking / Evaluation	Baseline implementation, ranking metrics, evaluation harness, initial OPE setup	TBD
Application / Systems	UI mockup, user-flow definition, system/tool interface design, conversational interaction design and implementation	TBD
Shared	Architecture decisions, documentation, integration, final documentation and demo	All

AI Engineering Focus Areas

Data

- Use an existing public dataset

Model and System

- Tool use or agentic orchestration

Application and Deployment

- Lightweight user interface

- Deployment option once finalized

Evaluation and Responsible AI

- Custom test suite beyond the baseline harness
- Robustness tests for edge cases
- Bias analysis, where applicable
- Safety and hallucination checks, where applicable
- Latency and cost profiling

Weekly Check-In

Progress Made

Following the feedback from the professor and TA, we revised the project while retaining the core retrieval, reranking, and OPE components from the original proposal. We selected a consumer-facing recommendation experience as the primary use case, added a user-facing interface and natural-language preference refinement to the planned system. A GitHub repository has also been created for the project.

Top Blockers / Risks

- Selecting the primary recommendation domain and dataset. Keeping the full system within the 14-week scope

Planned Next Steps

- Evaluate the candidate datasets and select the primary domain/dataset.
- Finalize the initial baselines and evaluation split.
- Define the initial user flow and UI mockup.
- Begin Milestone 2 related work.
- Define team cadence.